

The Differential Equation $\dot{x} = f \circ x$

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A theory is presented for absolutely continuous solutions of the general scalar first order autonomous o.d.e. Necessary and sufficient conditions are given for local and global existence and uniqueness, and further topics include existence-uniqueness duality, structure of the solution set and weak solutions.

1. INTRODUCTION

Absolutely continuous solutions x will be studied here for

$$\dot{x}(t) = f \circ x(t), \quad t \geq 0 = x(0). \quad (1.1)$$

Many of the standard questions, like existence and uniqueness, can be analysed for this equation without highly sophisticated mathematics. On the other hand, the problem is not altogether trivial and various counterexamples (some new for *any* o.d.e.) will be given using the form (1.1).

When $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous, x is $C^{(1)}$ and the corresponding existence question was completely solved by Peano [14] in 1885: (1.1) has a (pointwise) minimal solution x given by

$$x(t) = \sup\{y(t): y(0) = 0, \dot{y}(s) < f \circ y(s) \forall s \in]0, t[\} \quad (1.2)$$

without any further conditions. The maximal solution is given by a dual formula, and in [15] Peano also noted that $f(\cdot) \neq 0$ is sufficient for unique solutions, since then

$$\int_{x(0)}^{x(t)} 1/f = t. \quad (1.3)$$

This idea was subsequently used by Osgood [13] for his well-known uniqueness criterion which essentially uses (1.1) as a comparison equation. Osgood allowed $f(0) = 0$ but required $f(\alpha) > 0$ whenever $\alpha > 0$. Then $1/f$ is (improperly) Riemann integrable, and the uniqueness criterion reduces to whether the

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integral in (1.3) converges. Further progress was made by Wallach [22] who permitted $f(\alpha) \geq 0$ when $\alpha \geq 0$, using Lebesgue integration. Wallach characterized all solutions of (1.1) and in particular showed that the maximal solution obeyed (1.3).

Most of the literature on non- $C^{(1)}$ solutions concerns non-autonomous right sides discontinuous in t . For example, de la Vallée Poussin [21] studied equations continuous in $x(t)$, but Riemann integrable in t , and subsequently Carathéodory generalised this to Lebesgue integrability. Recently Goodman [6] has shown that the maximal solution of the scalar Carathéodory o.d.e. obeys (1.2) with nonstrict inequality a.e.s. In the autonomous case, of course, such right sides are continuous, but it will be shown in Section 7 that Goodman's result holds in general for (1.1), provided that solutions exist.

The necessary and sufficient conditions for local existence are obtained in Sections 3 and 4, and require $1/f$ to be Lebesgue integrable (when f is integrable, so is $\dot{x}^2$, and conversely) and to obey a monotonicity condition that avoids *jamming*. The sgn function obeying

$$\text{sgn } \alpha = -1 \quad \text{if } \alpha < 0, \quad = 1 \quad \text{if } \alpha > 0 \quad (1.4)$$

does not jam, but $-\text{sgn}$ jams at 0. When $f = \text{sgn}$, (1.1) has nonunique solutions; necessary and sufficient conditions for uniqueness are given in Section 5. Further, it is well known that uniqueness of solutions implies their continuous dependence on $x(0)$ if f is continuous, but the converse question has been open for some time. An example with continuous f is given in Section 5 where the maximal and minimal solutions are distinct but continuously dependent on $x(0)$.

When $f = -\text{sgn}$ (and $\text{sgn } 0 \neq 0$), (1.1) has no solutions, illustrating a duality between local existence and uniqueness. Global duality (for continuous f) has been studied at various levels of detail by Wintner, Brauer–Sternberg [3] and Yoshizawa [24]. Both local and global dualities are discussed for the general case in Section 6, along with global existence. In addition to the usual possibilities, a solution x to (1.1) may exist on $[0, \omega]$ but simply *stop* at finite $x(\omega)$.

Section 7 contains characterizations of various extremal solutions with analogues of (1.2), together with conditions for their existence, and the structure of the solution set is deduced. When f is continuous, the set of all possible $x(t)$ (for fixed t) is a closed interval, another result of Peano [14]. Here there may be three intervals; only one need be closed and all could be empty.

Finally a brief discussion of “weak” solutions is given in Section 8. Two standard types of definition, one involving a convex set valued right side, the other involving limits of “nearby” equations, are examined. The first definition, based by Rosenthal [18] on earlier work of Nagumo [11] and since investigated e.g. by Filippov [5], is shown to guarantee solutions iff f does not take too many infinite values. The second definition turns out to be more restrictive than the first, but is weaker (and hopefully more realistic) than other “approximation” definitions that I have seen.

Before we proceed, it should be pointed out that most of the material here extends without great difficulty to the separable equation

$$\dot{x}(t) = f(x(t))g(t). \quad (1.5)$$

A special discontinuous case of this was considered in de la Vallée Poussin's memoir [21] referred to earlier, and a more substantial investigation (for continuous f and g) was made by Kamke [8] with a view to using (1.5) as a comparison equation for uniqueness conditions. Also Olech-Plis [12] have shown that if (1.5) satisfies Perron's uniqueness criterion [16] then Picard's iterates converge to a solution. (Uniqueness alone is not enough even for (1.1), as $f(x) = -x^{1/3}$ shows; there are of course well-known nonautonomous examples of this phenomenon.)

2. PRELIMINARIES

Absolutely continuous and bounded variation will be abbreviated to AC and BV throughout, and the symbol x will always denote an AC solution to (1.1) on a nondegenerate interval $J = [0, \omega]$. Lebesgue measure is denoted by λ and "Lebesgue" will be omitted from associated concepts. Also the standard contractions

$$\int g(t) d\lambda(t) = \int g(t) dt = \int g$$

will be used. We shall need three results at various points, the first two being well known.

LEMMA 2.1 [17; IV.1.4 Cor. 2]. *If $S \subseteq J$ is measurable and y is AC on J then $y(S)$ is measurable.*

LEMMA 2.2 [17; V.1.6 Th. 4]. *If y is continuous and BV on J , then y is AC iff y preserves null sets, i.e. $N \subseteq J$ and $\lambda(N) = 0 \Rightarrow \lambda(y(N)) = 0$.*

LEMMA 2.3. *Let y be AC on J , g be measurable (perhaps infinite valued) on $y(J)$ and $h(t) = g(y(t))\dot{y}(t)$. Let S be a measurable subset of J . Then*

$$\int_{y(S)} g = \int_S h$$

provided that at least one of the following holds:

- (i) g is integrable over $y(J)$, y increases and S is an interval or
- (ii) h is integrable on J , g is finite valued and S is an interval or
- (iii) h is integrable on J , g is finite valued and y increases.

This is a slight generalization of the standard change of variable rule. The proof of (i) is given in [10; Th. 38.4] and (ii) in [17; V. 1.6 Th. 7]; the proof of (iii) can be carried out as follows.

Observe that y is constant at most on a countable set C of nontrivial intervals, and that if y^i denotes the greatest inverse function of y then $y^i \circ y$ is the identity on $J \sim C$. Let χ be the characteristic function of S and note that $\psi = \chi \circ y^i$ is measurable since $\psi^{-1}(\{1\}) = y(S)$ is measurable by Lemma 2.1. Now $\dot{y}$ vanishes outside $J \sim C$ while $\psi \circ y$ and $g \circ y$ are both finite valued. Thus

$$\int_J \psi(y(s)) g(y(s)) \dot{y}(s) ds = \int_{J \sim C} \psi(y(s)) h(s) ds \quad (2.1)$$

if either side exists. But the integral on the right is dominated by h (integrable by assumption) and, on $J \sim C$, $\psi \circ y = \chi$ (measurable by assumption). Thus both integrals exist and equal $\int_{S \sim C} h = \int_S h$, using finiteness of g again.

Finally by (ii), the integral on the left of (2.1) is

$$\int_{y(J)} \psi(\alpha) g(\alpha) d\alpha = \int_{y(S)} g,$$

completing the proof.

3. NECESSARY CONDITIONS FOR LOCAL EXISTENCE

There are basically three conditions on f : measurability, lack of jamming, and reciprocal integrability. The first follows from Lemma 2.1.

LEMMA 3.1. *If x exists then f is measurable on $x(J)$.*

Proof. Let $P \subseteq f \circ x(J)$ be open, so

$$\dot{x}^{-1}(P) = x^{-1}(f^{-1}(P))$$

is a measurable set. By virtue of Lemma 2.1, $f^{-1}(P) = x(\dot{x}^{-1}(P))$ is therefore measurable, Q.E.D.

The next preliminary result was given in [22; Sect. 2] for continuous f .

LEMMA 3.2. *x is monotonic.*

Proof. Suppose there are $t_i \in J$ with $x(t_1) = x(t_2) = \alpha$, say, and

$$\beta = \sup\{x(t): t_1 \leq t \leq t_2\} > \alpha. \quad (3.1)$$

Let the supremum be attained at $t = s$, and set

$$\sigma = \sup\{t < s: x(t) = \alpha\}, \quad \tau = \inf\{t > s: x(t) = \alpha\}.$$

Then $\sigma < s < t$, $x(\sigma) = x(\tau) = \alpha < \beta = x(s)$, and

$$t \in [\sigma, \tau] \Rightarrow \alpha \leq x(t) \leq \beta.$$

Now consider Lemma 2.3(ii) with $y = x$ and $g = \operatorname{sgn} f$ where sgn satisfies (1.4) and $\operatorname{sgn} 0 = 0$. Then Lemma 3.1 shows that g is measurable, while

$$h: t \rightarrow \operatorname{sgn} f(x(t)) \dot{x}(t) = |\dot{x}(t)|$$

is integrable because x is AC . Applying Lemma 2.3(ii),

$$\int_{\alpha}^{\beta} \operatorname{sgn} f = \int_{\sigma}^s |\dot{x}(t)| dt, \quad \int_{\beta}^{\alpha} \operatorname{sgn} f = \int_s^{\tau} |\dot{x}(t)| dt.$$

Adding, we obtain $\dot{x} = 0$ a.e. on $[\sigma, \tau]$, contradicting (3.1). There is a similar argument countering

$$\inf\{x(t): t_1 \leq t \leq t_2\} < \alpha, \quad \text{Q.E.D.}$$

In view of this last result, nonzero solutions to (1.1) are either positive or negative for $t > 0$. Thus we shall frequently assume $x(t) \geq 0$ or > 0 for $t > 0$, the general case just being clumsier.

We come now to the key integral formula, satisfied by all nonnegative solutions x .

THEOREM 3.3. $\lambda([s, t] \sim \dot{x}^{-1}(0)) = \int_{x(s)}^{x(t)} 1/f$ whenever $s, t \in J$.

Proof. We apply Lemma 2.3(iii) to $y = x$ (increasing by Lemma 3.2) and $g = 1/f$ (measurable by Lemma 3.1). Evidently $h \equiv 1$ and

$$\lambda([s, t] \sim \dot{x}^{-1}(0)) = \int_{[x(s), x(t)] \sim f^{-1}(0)} 1/f$$

so it suffices to show that $S = f^{-1}(0) \cap x(J)$ is a null set. Evidently $x^{-1}(S) = \dot{x}^{-1}(0)$ is measurable, so using Lemma 2.3(iii) again with $y = x$ and $g \equiv 1$,

$$\lambda(S) = \int_{x^{-1}(S)} \dot{x} = 0, \quad \text{Q.E.D.}$$

The remaining necessary conditions now follow easily. First, we need to ensure that trajectories do not jam together.

DEFINITION 3.4. f is almost nonnegative over $]0, \beta]$ (abbreviated to $an\beta$) if $f(\alpha) \geq 0$ for almost every $\alpha \in]0, \beta]$.

The following notation will also be useful:

$$f_{\gamma+}(\alpha) = f(\gamma + \alpha), \quad f_{\gamma-}(\alpha) = f(\gamma - \alpha); \quad (3.2)$$

$$Q = \{\delta \in x(J) : |f(\delta)| = \infty\}. \quad (3.3)$$

We can now formulate the necessary condition theorem in the above notation.

THEOREM 3.5. *If (1.1) has a solution then $\lambda(Q) = 0$, and either*

- (i) $f(0) = 0$, or
- (ii) $\exists \beta > 0$ so that f is $an\beta$ and $\int_0^\beta 1/f$ exists, or
- (iii) as per (ii) but replacing f by f_{0-} .

Proof. If $\lambda(Q) > 0$ then Lemma 2.2 gives $\lambda(x^{-1}(Q)) > 0$. Thus

$$\{t : |\dot{x}(t)| = |f \circ x(t)| = \infty\}$$

also has positive measure, contradicting integrability of $\dot{x}$.

Next, $f(0) = 0$ trivially allows $x = 0$ as a solution to (1.1), so without loss assume $f(0) \neq 0 < \xi = x(\omega)$. From Theorem 3.3, $1/f$ is integrable over $[0, \xi]$ and by Lemma 3.2, $\int_0^\delta 1/f$ increases with δ . Consequently f is nonnegative valued a.e. on $[0, \xi]$. Q.E.D.

With f continuous and $x \geq 0$ imposed, these conditions reduce to $f \geq 0$ and $1/f$ integrable, as given by Wallach [22].

4. SUFFICIENT CONDITIONS FOR LOCAL EXISTENCE

The principal result of the section is that the conditions of Theorem 3.5 are in fact sufficient as well.

THEOREM 4.1. *If $f(0) = 0$ or $\exists \beta > 0$ so that f is $an\beta$ with $1/f$ integrable over $[0, \beta]$ and $Q = \{\delta \in [0, \beta] : |f(\delta)| = \infty\}$ is null, then (1.1) has a nonnegative solution.*

Observe that these hypotheses guarantee measurability of f , the remaining necessary condition of Lemma 3.1. Nonpositive solutions are generated by using f_{0-} (3.2) instead of f .

Proof. Obviously $f(0) = 0$ is trivial, so let $q: \alpha \rightarrow \int_0^\alpha 1/f$ exist on $[0, \beta]$. Since Q is null, q increases strictly and continuously, so

$$v = q^{-1}: J = [0, q(\beta)] \rightarrow [0, \beta]$$

increases (hence is BV), is continuous, and satisfies

$$\int_0^{v(t)} 1/f = t. \quad (4.1)$$

Let $N \subseteq J$ be null and χ be the characteristic function of N . Thus Lemma 2.3(i), with $g = \chi$, $y = q$ gives

$$\int_0^\gamma \chi(q(\alpha)) q'(\alpha) d\alpha = \int_0^{q(\gamma)} \chi = 0$$

for all $\gamma \in q(J)$. Now Q is null so $q'(\alpha) = 1/f(\alpha) \neq 0$ a.e. $\alpha \in q(J)$ whence $\chi \circ q$ vanishes a.e., proving that $q^{-1}(N) = v(N)$ is null.

Thus v is AC by Lemma 2.2, and the chain rule [10; Th. 31.10] gives

$$[f \circ v(t)]^{-1} \dot{v}(t) = (q \circ v)'(t) = 1 \quad (4.2)$$

for almost all $t \in J \sim v^{-1}(S)$, S being the null set where q fails to be differentiable. Finally q is AC so $v^{-1}(S) = q(S)$ is null by Lemma 2.2, and (4.2) holds a.e. on J . Q.E.D.

Sufficiency was established by Peano [14; 3^o] for continuous f , and indicated by Azbelev-Tsalyuk [1] for increasing f (via the Knaster-Tarski fixed point theorem). Also Wend [23; Th. 2.1] has shown that Picard's iterates converge to a solution of (1.1) when f is increasing and nonnegative valued.

As is well known, x is an AC solution iff $\dot{x}$ is integrable.

COROLLARY 4.2. $\dot{x}^2$ is integrable iff f satisfies Theorem 3.5 and is integrable over $x(J)$.

Proof. Let x satisfy (1.1) and increase, without loss of generality. For any $t \in J$, if f is integrable then Theorem 4.1 proves existence of x , and Lemma 2.3(i) gives

$$\int_0^t \dot{x}^2 = \int_0^t f(x(s)) \dot{x}(s) ds = \int_0^{x(t)} f, \quad (4.3)$$

so $\dot{x}^2$ is integrable. Conversely if $\dot{x}^2$ is integrable then (4.3) holds at least where the integrands are finite by Lemma 2.3(ii). Thus (4.3) holds, since Q and $x^{-1}(Q) = \dot{x}^{-1}(\infty)$ are both null, so f is integrable.

The following shows that absolute continuity of x does not imply integrability of f .

EXAMPLE 4.3. Let $f(\alpha) = \alpha^{-1}$, so $x(t) = (2t)^{1/2}$ satisfies (1.1).

5. UNIQUENESS AND CONTINUOUS DEPENDENCE

From Theorem 3.5 we can deduce a necessary and sufficient condition for uniqueness of solutions. This will also admit a useful interpretation in terms of duality via solution reversal, as will be seen in Section 6. We need some definitions first, negating the local existence conditions: see Definition 3.4 and Theorem 3.5.

DEFINITION 5.1. f jams (from above) if f is $\alpha\beta$ for no $\beta > 0$.

DEFINITION 5.2. f attracts (from above) if $\int_0^\beta 1/f$ exists for no $\beta > 0$.

The corresponding notions from below would be phrased in terms of f_{0-} (3.2), but for simplicity we shall consider only *nonnegative* solutions, and these definitions will suffice.

If $f(0) = 0$ then we deny non-zero solutions by negating Theorem 3.5(ii) and (iii). If $f(0) \neq 0$ and a positive solution x exists then it is enough to ensure that from no $x(t)$ can solutions bifurcate by continuing via $x(s) = x(t)$ as well as $x(s) > x(t)$ for $s > t$. We state this formally.

THEOREM 5.3. x is unique iff $f(\alpha) = 0 \Rightarrow \exists \alpha \in x(J) \Rightarrow f_{\alpha+}$ jams and attracts.

With f continuous, several authors have contributed to this question. Peano gives the sufficient condition that f should never vanish ([15; p. 227], the vacuous case of Th. 5.3), while Osgood [13; p. 344] assumes that f attracts and $f(\alpha) > 0$ when $\alpha > 0$. Necessary and sufficient conditions were given by Kamke [8; Sect. 4], jamming being replaced by nonvanishing on $]0, \beta]$. Wend [23; Th. 2.2] has shown that $t > 0 \Rightarrow f \circ x(t) > 0$ for all solutions x suffices for uniqueness: this holds regardless of continuity of f . As a corollary, $f(0) > 0$ suffices when f increases [23; Cor. 2.3].

Assume now that f is *continuous*. It is a well-known result (cf. [19; Cor. 3.1]) that solutions depend upper semicontinuously on initial data: in particular, uniqueness implies continuous dependence. The converse question [19; p. 385], whether continuous dependence in Hausdorff's metric implies uniqueness, seems open still, and the next example answers it negatively.

EXAMPLE 5.4. Let $f(\alpha) = 2|\alpha|^{1/2}$ for $\alpha \leq 0$, $\theta_n = 2^{-n}\pi^{-1}$ and $f(\alpha) = 0$ if $\alpha \geq \theta_1$. For $\alpha \in]0, \theta_1]$, f is defined by

$$f(\alpha) = 2[(\alpha - \theta_n)(\theta_{n-1} - \alpha)/\theta_n]^{1/2} \quad \text{if } \theta_n \leq \alpha \leq \theta_{n-1}. \quad (5.1)$$

Evidently the maximum value of f in (5.1) occurs at $\alpha = \frac{3}{2}\theta_n$:

$$0 \leq f(\alpha) \leq \theta_n^{1/2} \quad \text{if } \theta_n \leq \alpha \leq \theta_{n-1}, \quad n > 1. \quad (5.2)$$

In particular, f is continuous. Thus [14; 50] for any initial value ϵ , the complete set of solutions forms an interval between the minimal and maximal solutions u_ϵ and v_ϵ respectively. It will now be shown that for all $t > 0$, $u_\epsilon(t)$ and $v_\epsilon(t)$ are distinct and depend continuously on ϵ at $\epsilon = 0$.

Since $f \geq 0$, minimal solutions satisfy

$$\begin{aligned} u_\epsilon(t) &= u_0(t) = 0 & \text{if } -t^2 \leq \epsilon \leq 0 \\ u_\epsilon(t) &= \theta_{n-1} & \text{if } \theta_n < \epsilon \leq \theta_{n-1} \text{ and } \pi^2 \epsilon \leq 4t^2. \end{aligned} \quad (5.3)$$

(5.3) can be shown as follows, since $u_\epsilon(t) > v_{\theta_n}(t)$ for t such that $u_\epsilon(t) < \theta_{n-1}$. Now (5.1) gives the maximal solution

$$v_{\theta_n}(s) = \theta_n[1 + \sin^2(s\theta_n^{-1/2})], \quad 0 \leq s \leq \frac{1}{2}\pi\theta_n^{1/2} \quad (5.4)$$

so $v_{\theta_n}(\frac{1}{2}\pi\theta_n^{1/2}) = \theta_{n-1}$ and $\pi\epsilon^{1/2} \leq 2t$ suffices for $u_\epsilon(t) = \theta_{n-1}$. Thus $u_\epsilon(t)$ is continuous at $\epsilon = 0$, for each fixed $t > 0$.

Turning to maximal solutions, $v_\epsilon(t) = v_0(t - |\epsilon|^{1/2})$ when $-t^2 \leq \epsilon \leq 0$. From (5.2), $v_0(\cdot)$ never exceeds $\theta_1^{1/2} < 1$, so $|v_\epsilon(t) - v_0(t)| < \epsilon^{1/2}$. When $\theta_n < \epsilon \leq \theta_{n-1}$

$$|v_0(t) - v_\epsilon(t)| \leq v_\epsilon(v_0^{-1}(\epsilon)) < v_0^{-1}(\epsilon)$$

again using $v_\epsilon(\cdot) < 1$. Using (5.4),

$$v_0^{-1}(\theta_{n-1}) = \sum_{r=n}^{\infty} \frac{1}{2}\pi\theta_0^{1/2} = \pi^{1/2}2^{-n/2}/(2 - 2^{1/2})$$

so

$$v_0^{-1}(\epsilon) \leq v_0^{-1}(\theta_{n-1}) = O(\theta_n^{1/2}) = O(\epsilon^{1/2})$$

and continuity of $v_\epsilon(t)$ at $\epsilon = 0$ has been demonstrated.

When f is *discontinuous*, there is of course no reason even to expect continuous dependence to follow from uniqueness, as the following shows.

EXAMPLE 5.5. $f(\alpha) = 1$ if $\alpha \geq 0$, $= 0$ if $\alpha < 0$.

6. CONTINUATION AND DUALITY

Throughout this section, we shall restrict attention to *nonnegative* solutions. Analogous results exist for the general case but gain only clumsiness. After a discussion of existence intervals for (1.1), we establish connections between continuability and reverse non-uniqueness. The one-sided nature of the problem is clearer here, for existence to the right bears no implications for uniqueness to the right (or existence to the left, in contrast with the case of continuous f).

The existence interval of the maximal (Th. 3.3) solution v (4.1) obeying

$$\int_0^{v(t)} 1/f = t \quad (6.1)$$

depends on the region of integrability of $1/f$. If $1/f$ is integrable over $[0, \xi[$ but not over $[0, \beta]$ when $\beta > \xi$ then there are four possibilities: in the first, ξ may be finite or infinite.

$$\lim_{\alpha \uparrow \xi} \int_0^\alpha 1/f = \infty, \quad \text{i.e.} \quad \int_0^\xi 1/f \text{ is divergent} \quad (6.2)$$

$$\lim_{\alpha \uparrow \xi} \int_0^\alpha 1/f = \omega < \infty, \quad f(\xi) \neq 0, \quad \xi < \infty, \quad (6.3)$$

$$\lim_{\alpha \uparrow \xi} \int_0^\alpha 1/f = \omega < \infty, \quad f(\xi) = 0, \quad \xi < \infty, \quad (6.4)$$

$$\xi = \infty, \quad \text{i.e.} \quad \int_0^\infty 1/f = \omega < \infty. \quad (6.5)$$

In (6.2), $v(t) \rightarrow \xi$ as $t \uparrow \infty$, so v is a solution over the half line $t \geq 0$. In (6.3), $v(t) \uparrow \xi$ as $t \uparrow \omega$ and v simply *stops* at $v(\omega) = \xi$. Continuation is impossible because Theorem 3.5 fails with f replaced by $f_{\xi+}$. In (6.4), we again have $v(\omega) = \xi$ but continuation is possible for $t \geq \omega$ via $v(t) \equiv \xi$. Finally in (6.5) we have the phenomenon of escape to infinity for at least one solution: $v(t) \uparrow \infty$ as $t \uparrow \omega$.

The duality theory is based on the elementary observation that if $f(0) = 0$ and the zero solution is not unique, then reversed solutions reach zero value from nearby. For any function g , let g^α coincide with g except that $g^\alpha(0) = \alpha$, ($\alpha = 0, 1$). The first result concerns local duality for (1.1)-recall (3.2) for the notation f_{0-} . The reversed equation corresponding to (1.1) is $\dot{x} = -f_{0-} \circ x$ but we shall use $+f_{0-}$ to give a nonnegative solution.

LEMMA 6.1. *The zero solution for f^0 is nonunique $\Leftrightarrow$ there is at least one solution for f^1 through zero $\Rightarrow$ the zero solution for f_{0-}^0 is unique $\Leftrightarrow$ there is no solution for f_{0-}^1 through zero.*

By "solution for g " is meant "solution of (1.1) with f replaced by g ". Lemma 6.1 is a straightforward application of Theorems 3.5 and 5.3.

Next consider the failure of global existence with $\xi < \infty$ i.e. case (6.3). Using Lemma 6.1 at (ω, ξ) instead of $(0, 0)$, we have the following.

COROLLARY 6.2. *If global existence fails for (1.1) via (6.3) then the zero solution for $f_{\xi+}^0$ is unique and that for $f_{\xi-}^0$ is nonunique.*

The converse holds if $x(0) \neq 0$ is permitted—specifically $\dot{x} = f \circ x$ fails global existence for some $x(0) < \xi$.

Finally we consider escape to infinity. In fact, Corollary 6.2 still holds for $\xi = \infty$ with suitable interpretation, but it is simpler to use

$$\bar{f}(\alpha) = \alpha^2 f(1/\alpha) \quad \text{if } \alpha > 0, \quad \bar{f}(0) = 0,$$

rather than to attach a meaning to f_{ξ^-} .

LEMMA 6.3. (i) *Global existence fails for f via (6.5) iff it fails for $\bar{f}$ via (6.5).*

(ii) *The zero solution for $\bar{f}$ is nonunique iff escape to infinity is possible for $\dot{x} = f \circ x$ from some $x(0) < \infty$.*

Proof. (i) Let (6.5) hold (for f). Then

$$\int_{1/\tau}^{1/\sigma} \frac{d\alpha}{f(\alpha)} = \int_{\sigma}^{\tau} \frac{d\alpha}{\alpha^2 f(1/\alpha)} \quad (6.6)$$

whenever $0 < \sigma < \tau < \infty$, using $y(t) = 1/t$, $g = 1/f$ and $[\sigma, \tau]$ instead of J in Lemma 2.3(i). Using (6.5) we see that (6.6) remains valid as $\sigma \downarrow 0$, $\tau \uparrow \infty$. Now set $\bar{x} = 1/x$: evidently $\dot{\bar{x}} = \bar{f} \circ \bar{x}$ and

$$\int_0^{\infty} 1/\bar{f} = \int_0^{\infty} \frac{d\alpha}{\alpha^2 f(1/\alpha)} < \infty$$

is precisely (6.5) for $\bar{f}$ instead of f .

The converse follows from reversing the roles of f and $\bar{f}$ —note that $f(\alpha) = \alpha^2 \bar{f}(1/\alpha)$.

(ii) This is a special case of (i) where $\sigma \downarrow 0$ but τ remains finite. $[0, \tau]$ is an existence interval for a positive solution for $\bar{f}$ iff x escapes to infinity with $\dot{x} = f \circ x$ and $x(0) = 1/\tau$, Q.E.D.

Wintner, one of the first to investigate global existence systematically, was clearly aware of global duality. Brauer–Sternberg [3; p. 425] observe that when f is continuous, global existence failure implies nonuniqueness at infinity because, in this case, (6.3) cannot occur. Actually, Brauer–Sternberg assert that if $\bar{x} = M(x)$ for a “smooth homeomorphism M which vanishes sufficiently rapidly at infinity” then (6.5) is equivalent to nonuniqueness of $\bar{x} = 0$. Yoshizawa [24; pp. 14–15] has given such a function M for continuous f (in fact [3] and [24] treated nonautonomous systems in $\mathbb{R}^n$ but M was simply a function of $\|x\|$). For (1.1), $M(x) = 1/x$ suffices for arbitrary f .

The following shows that $x(0) = 0$ is not always possible in (ii) even when f is continuous.

EXAMPLE 6.4. Let $f(\alpha) = (\alpha - 1)^2$ so if $x(0) = 0$ then $x(t) = 1 - (t + 1)^{-1}$ i.e. case (6.2) with $\xi = 1$. On the other hand x will escape to infinity from $x(0) > 1$ since then

$$x(t) = 1 + (x(0) - 1 - t)^{-1}.$$

Observe that $\bar{f} = f^0$ in this case, so $\bar{x}(t) = 1 - (t + 1)^{-1}$ is a nonzero solution with $\bar{x}(0) = 0$.

7. STRUCTURE OF THE SOLUTION SET

DEFINITION 7.1. The t -isochrone $I_t = \{x(t): x \text{ satisfies (1.1)}\}$.

We shall assume that all solutions are continuable to t . When f is continuous, I_t is a nonempty closed interval [14; 50] and the classical results of Kneser on connectedness and Hukuhara on peripheral attainability follow immediately. All these results fail for the general case of (1.1), as the following examples show.

EXAMPLE 7.2. Let $f^*(\alpha) = 0$ if $\alpha = 1, \frac{1}{2}, \frac{1}{3}, \dots$, $f(\alpha) = 1$ otherwise. Then $v(t) = t$ gives the maximal solution v through the origin, but there is *no* minimal solution. When $t > 0$, $I_t =]0, t]$.

EXAMPLE 7.3. Let

$$f(\alpha) = \operatorname{sgn} \alpha (\operatorname{sgn} 0 = 0) \quad \text{for } |\alpha| \leq 1,$$

$$= f^*(\alpha - 1) \quad \text{for } \alpha > 1$$

and

$$= -f^*(-\alpha - 1) \quad \text{for } \alpha < -1.$$

When $t > 1$, I_t consists of three disjoint intervals:

$$I_t = [-t, -1[\cup \{0\} \cup]1, t].$$

EXAMPLE 7.4. Let $f(0) = 1$, $f(\alpha) = 0$ if $\alpha \neq 0$. Then I_t is empty for all $t > 0$.

The above examples are in a sense as bad as possible, and more positive results will now be given. We assume for simplicity that all solutions are *positive* for $t > 0$. Recall the definition of v (4.1).

LEMMA 7.5. *Every solution of (1.1) can be represented as the greatest inverse of $v + x_1 + x_2$ where x_i are constant except on $f^{-1}(0)$, x_1 is continuous and strictly increasing on $f^{-1}(0)$, and x_2 is an increasing step function.*

Wallach's proof [22; Sect. 7] for continuous f carries over because of Lemma 3.2. The next result shows that some extremal solutions exist if any solutions do so, and that an appropriate analogue of (1.2) also holds.

LEMMA 7.6. (i) $v = \sup\{y \in AC: y(0) = 0, \dot{y} \leq f \circ y \text{ a.e.}\}$

(ii) $v = \inf\{z \in AC: z(0) = 0, \dot{z} > f \circ z \text{ a.e.}\} \text{ iff } f \geq 0 \text{ on } v([0, \omega]).$

Here the order is pointwise (unless a.e. is specified).

Proof. (i) Without loss we can restrict the subfunctions y to increasing functions so

$$t \geq t - \lambda(\dot{y}^{-1}(0)) \geq \int_0^{y(t)} 1/f$$

follows as for Th. 3.3. Thus $y(t) \leq v(t)$.

(ii) If $f \circ v(s) < 0$ for some $s \in [0, \omega]$, choose z as the maximal solution for g , where $g(\alpha) = f(\alpha) - f \circ v(s)$, until $z(t) = v(s)$. Then continue z as $v(s)$ over $[z^{-1}(v(s)), \omega]$. Clearly $z(0) = 0$, $\dot{z} > f \circ z$ a.e. yet $z(t) < v(t)$ for $t > s$. Conversely, if $f \geq 0$ on $v(J)$ then $\dot{z}$ is positive a.e. and we apply the argument used above for y to give

$$t = t - \lambda(\dot{z}^{-1}(0)) < \int_0^{z(t)} 1/f$$

whence $z(t) > v(t)$. Finally it is clear that

$$z_\epsilon: t \rightarrow v(t + \epsilon t)$$

approach v (pointwise) as $\epsilon \downarrow 0$ and satisfy $z_\epsilon(0) = 0$, $\dot{z}_\epsilon > f \circ z_\epsilon$ a.e., Q.E.D.

With f continuous, (ii) was proved by Peano [14; 3^o]; see also [16] and [6]. With f increasing, (i) follows from the Knaster-Tarski fixed point theorem. Lemma 7.5 shows that if a solution x is given to (1.1) then smaller solutions may be obtained by adding constant segments at values $\in f^{-1}(0)$. Thus a minimal solution u will exist provided that the minimal positive element of $f^{-1}(0)$ is isolated, and then u equals v on part of J and is constant on the remainder. Minor amendments to the argument of Lemma 7.6 now prove the following.

LEMMA 7.7. Let $\alpha = \min\{\beta > 0: f(\beta) = 0\} < \inf\{\beta > \alpha: f(\beta) = 0\}$. Then the minimal solution of (1.1) exists and

(i) $u = \sup\{y \in AC: y(0) = 0, \dot{y} < f \circ y \text{ a.e.}\}$

(ii) $u = \inf\{z \in AC: z(0) = 0, \dot{z} \geq f \circ z \text{ a.e.}\} \text{ iff } f > 0 \text{ on } u(J).$

8. WEAK SOLUTIONS

Evidently the conditions of Theorem 3.5 do not always hold, even when f is measurable. Examples from physics and from the calculus of variations (with extremal fields that jam according to Def. 5.1) were studied even in the last century. The question of what solution definition will permit local existence under jamming conditions has been tackled by many authors, e.g. Nagumo [11] for o.d.e. and Bogoliubov [2] for the calculus of variations. These belong to the class of definitions in 8.2 below: for local existence we shall restrict t to $J = [0, \omega]$ and $x(t)$ to $K = [-\beta, \beta]$.

DEFINITION 8.1. Let Σ be a class of subsets of K . The Σ upper limit ${}^{\Sigma}f(\alpha) = \inf\{\sup\{f(\gamma): |\gamma - \alpha| \leq \epsilon, \gamma \notin S\}: \epsilon > 0, S \in \Sigma\}$.

In particular, if Σ consists of the empty set (or all finite subsets) then ${}^{\Sigma}f$ is $\overline{\lim}_{\gamma \rightarrow \alpha} f(\gamma)$, while if Σ consists of all null sets then ${}^{\Sigma}f$ is the "essential" limit superior. The lower limit ${}_{\Sigma}f$ is defined similarly.

DEFINITION 8.2. x is a Σ weak solution of (1.1) if x is AC and

$${}_{\Sigma}f \circ x \leq \dot{x} \leq {}^{\Sigma}f \circ x \quad \text{a.e. on } J.$$

This amounts to taking the convex all of f values at nearby points (excepting Σ), thus forming a set valued function

$$F_{\Sigma}: \alpha \rightarrow [{}_{\Sigma}f(\alpha), {}^{\Sigma}f(\alpha)] \quad (8.1)$$

and replacing (1.1) by the differential selection $\dot{x} \in F_{\Sigma} \circ x$. The works of Nagumo and Bogoliubov effectively take finite subsets for Σ , while that of Rosenthal [18] and the well known paper of Filippov [5] take Σ as all the null sets. (Actually [11] and [18] use Dini derivatives instead of $\dot{x}$, x being continuous, so AC solutions coincide with those of Def. 8.2). Let $f_{\Sigma} = {}_{\Sigma}f$ where this is positive, $= {}^{\Sigma}f$ where negative, and $= 0$ elsewhere.

THEOREM 8.3. Let all sets in Σ be null. Then (1.1) has a Σ weak solution x iff

$$\{\alpha \in x(J): |f_{\Sigma}(\alpha)| = \infty\} \quad (8.2)$$

is null.

Proof. Observe that ${}_{\Sigma}f$ and ${}^{\Sigma}f$ are measurable since semicontinuous. Thus f_{Σ} is measurable and, if (8.2) holds, satisfies the conditions of Theorem 4.1. Since $f_{\Sigma}(\alpha) \in F_{\Sigma}(\alpha) \forall \alpha \in K$, sufficiency is proved.

Conversely, if x is a Σ weak solution of (1.1) then $|f \circ x(t)| \geq |f_{\Sigma} \circ x(t)|$ for all $t \in J$ by construction. If (8.2) fails then $|\dot{x}(t)| = \infty$ on a set of positive measure, contradicting integrability of $\dot{x}$, Q.E.D.

Both Nagumo [11] and (in the case of (1.1)) Rosenthal [18] proved existence of their solutions for bounded f . In this case, the solutions are then AC [18; p. 4], hence follow Definition 8.2. In particular, when Σ includes *all* null sets, as in [18], solutions are guaranteed subject to the minimal requirement (8.2). This definition can of course cut out Nagumo solutions, or even ordinary (strong) solutions, as in the following.

EXAMPLE 8.4. $f(\alpha) = 1$ if $\alpha \neq 0$, $f(0) = 0$.

The zero solution of Example 8.4 is neither a Nagumo nor Rosenthal solution. The zero function is a Nagumo solution of Example 7.2 but neither a strong nor Rosenthal solution. It is also possible to allow certain sets of positive measure in Σ , but this will not be pursued.

Another type of solution definition arises when (1.1) is viewed as an approximate model for a moving physical object. Examples are relay chatter, water-pipe hammer and stick-slip elastic sliding, where the simplest models do jam. From the point of view of approximating a large class of nonlinearities frequently occurring in engineering and physics by discontinuities, it seems appropriate to consider graph (rather than uniform) approximations as follows.

Let G and H map $K \rightarrow 2^{\mathbb{R}}$, functions $K \rightarrow \mathbb{R}$ being viewed as singleton valued.

DEFINITION 8.5. G is an ϵ -approximation of H if the graph of H lies in an ϵ -neighbourhood of the graph of G , i.e.

$$\forall \alpha \in K \forall \gamma \in H(\alpha) \exists \delta \in K \exists \eta \in G(\delta): |\alpha - \delta| + |\gamma - \eta| \leq \epsilon.$$

It follows, for example, that the Sgn function (with $\text{Sgn } 0 = [-1, 1]$) is an ϵ -approximation of various hysteretic, delay, lag, etc. nonlinearities that occur in actual switch components, for suitable small ϵ . We now show in particular that limiting solutions, as the "imperfections" tend to zero, are weak solutions of (1.1). For simplicity we assume that f is bounded on K .

THEOREM 8.6. If F_{Σ} (8.1) is an ϵ -approximation to f^{ϵ} , and x^{ϵ} satisfy (1.1) with f^{ϵ} replacing f , then x^{ϵ} cluster (pointwise as $\epsilon \downarrow 0$) at Σ weak solutions to (1.1).

Proof. Since f^{ϵ} are uniformly bounded, x^{ϵ} are weakly L_1 sequentially compact on J [4; IV.8.11] so let x^{ϵ} cluster weakly in L_1 at $\hat{x}$. Evidently x^{ϵ} then cluster pointwise at x (by weak convergence and the fact that $x^{\epsilon}(0) = 0$). Further by Mazur's theorem [4; V.3.14], $\hat{x}(t)$ belongs to the convex limit

$$[\varliminf_{\epsilon \downarrow 0} f^{\epsilon} \circ x^{\epsilon}(t), \varlimsup_{\epsilon \downarrow 0} f^{\epsilon} \circ x^{\epsilon}(t)]$$

a.e. $t \in J$. Applying Definition 8.5 and using upper semicontinuity of F_{Σ} , we see that $\hat{x}(t) \in F_{\Sigma} \circ x(t)$, Q.E.D.

This result was proved for Nagumo solutions x , assuming $f^\epsilon \circ x^\epsilon \rightarrow f \circ x$ on J (e.g. if $f^\epsilon \rightarrow f$ uniformly on K), by Matrosov [9; Th. 1.3]. Since the crucial problem in applying Theorem 8.6 is existence of x^ϵ (requiring “nice” f^ϵ), it makes sense to relax the mode of approximation if possible—note that the Sgn function is not uniformly approximated by any of the “imperfect” characteristics mentioned earlier. Matrosov also gives a survey of other solution definitions; for a similar discussion of weak solutions for the calculus of variations, see [7].

In conclusion, it should be noted that the cluster solutions of x^ϵ may depend on the type of “imperfection” which tends to zero.

EXAMPLE 8.7.

$$\begin{aligned} f^\epsilon(\alpha) &= -1 \text{ if } \alpha < 0, = 1 \text{ if } \alpha > 2\epsilon \\ &= -1 + \alpha/\epsilon \text{ if } 0 \leq \alpha \leq 2\epsilon. \end{aligned}$$

Obviously Sgn is a 2ϵ -approximation to f^ϵ and to $f_{2\epsilon}^\epsilon$ (3.2) but the limiting solution to (1.1) for f^ϵ is $x(t) = -t$, and for $f_{2\epsilon}^\epsilon$ is $x(t) = +t$.

This has some relation to Utkin’s work, where

$$\dot{x} = f \circ g \circ x \tag{8.3}$$

is studied, with g discontinuous but approximating “imperfect” switch functions g^ϵ . Even with a constraint imposed so that Rosenthal solutions of (8.3) are unique, Utkin [20; Sect. 3] gives special types of approximation g^ϵ with different limits of solutions to (8.3) as g^ϵ tends to g . He also shows [20; Sect. 2], subject to the same uniqueness constraint, that these limiting solutions are unique if f is affine; this is a simple consequence of the proof of Theorem 8.6.

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